HAT-P-36b

R-0864

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_240315` · created 2026-07-28T22:30:14+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460384.7693 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.046
Transit depth (Rp/Rs)^2	2.43 +/- 1.44 [%]
Orbital Inclination (inc)	80.54 +/- 2.87
Airmass coefficient 1 (a1)	0.913 +/- 0.081
Airmass coefficient 2 (a2)	0.087 +/- 0.085
Scatter in the residuals of the lightcurve fit is	8.96 %
Best Comparison Star	#3 - [231, 396]
Optimal Aperture	1.85
Optimal Annulus	4.92
Transit Duration (day)	0.074 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.046 vs 0.126 (deviation 0.58σ)
Duration fit vs published	0.074 d vs 0.0925 d (deviation 0.83σ)
Mid-time O–C	-12.9 min
Depth SNR	3.0
Residual scatter	8.96 %

Light curve

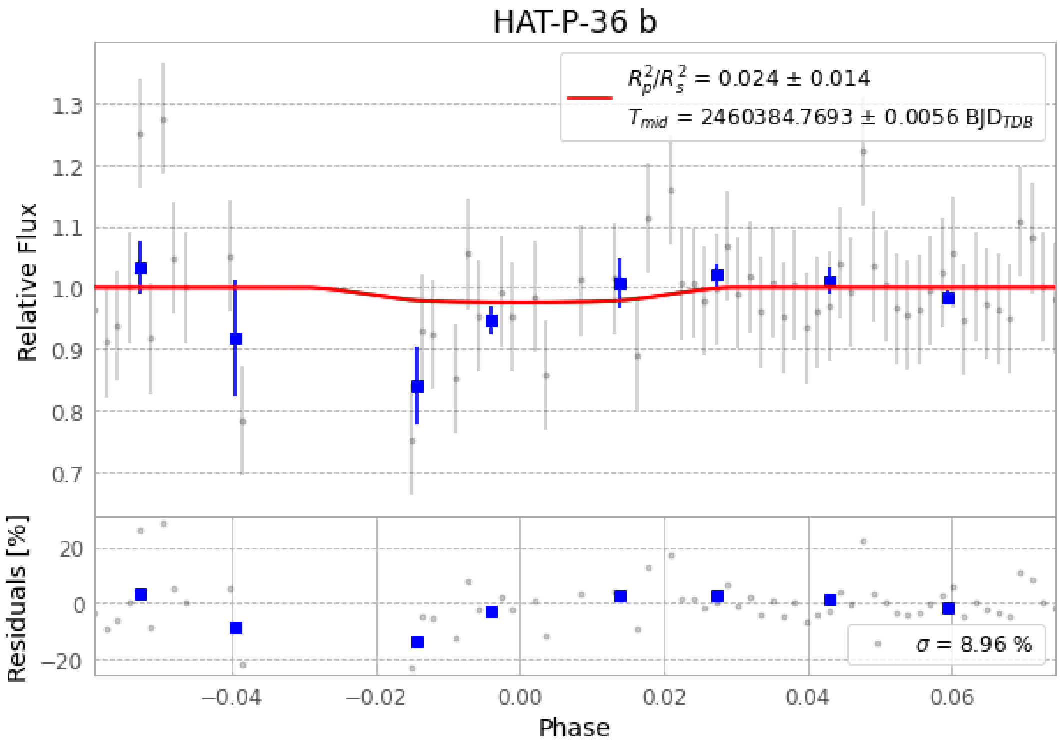

Figure 1 — FinalLightCurve_HAT-P-36 b_2024-03-14.png (SHA-256 1de80c1b7bd93289…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 250], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ddcdfba0db4ae95c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

86 FITS frames (56 MB), `HATP-36240315042715.FITS` → `HATP-36240315084215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240315.log (6c7e8cbf94da…), FinalLightCurve_HAT-P-36 b_2024-03-14.png (1de80c1b7bd9…), FinalLightCurve_HAT-P-36 b_2024-03-14.csv (baeeb71f4e9d…)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)